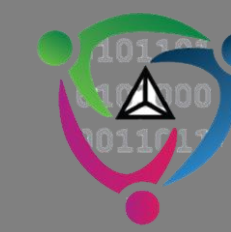


Toward a Structure-to-Function Science of Foundation Models



Center for Complex Particle Systems
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ACM AI Leadership Summit 2026 **Vision Paper**

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Our vision

Capabilities and intelligence emerge from organized interactions, not isolated components.
From structural organization to prediction, intervention, and inverse design.

Quantifying Structural Organization

Component	Neurons Attention heads MLP blocks Hidden states
Representation	Directions Vectors Sparse features
Circuits & traces	Task-specific Sparse-feature Attribution graphs
Structural Organization	
Behavior	Accuracy Refusal Robustness Jailbreak

Structure is the organizational layer through which local mechanisms become system-level functions.

Model-derived Structural Graphs

- **Choose units**
Neurons, heads, MLP blocks, sparse features
- **Declare a context**
Prompt, task family, behavioral condition, etc.
- **Measure and aggregate relations**
Measurement protocols applied to samples

$$A_{t,c,ij}^{(P)} = \text{Agg}_{x \sim c} [\rho_{ij}^{(P)}(x; M_t)] \quad (1)$$

$$G_{t,c}^{(P)} = (U, A_{t,c}^{(P)}, R, D) \quad (2)$$

U internal units · A aggregated relations · R representation class (e.g. directed weighted graph) · D metadata: unit scale, context, protocol, intended use.

A Menu of Relations as Edges

Relations	Captures
Parameter coupling	Static influence
Attention patterns	In-context token interaction
Activation statistics	Representational co-variation
Gradient / attribution	Local sensitivity
Intervention response	Causal influence
Information flow	Compression, transfer, bottlenecks

Overview: Structure-to-Function Framework

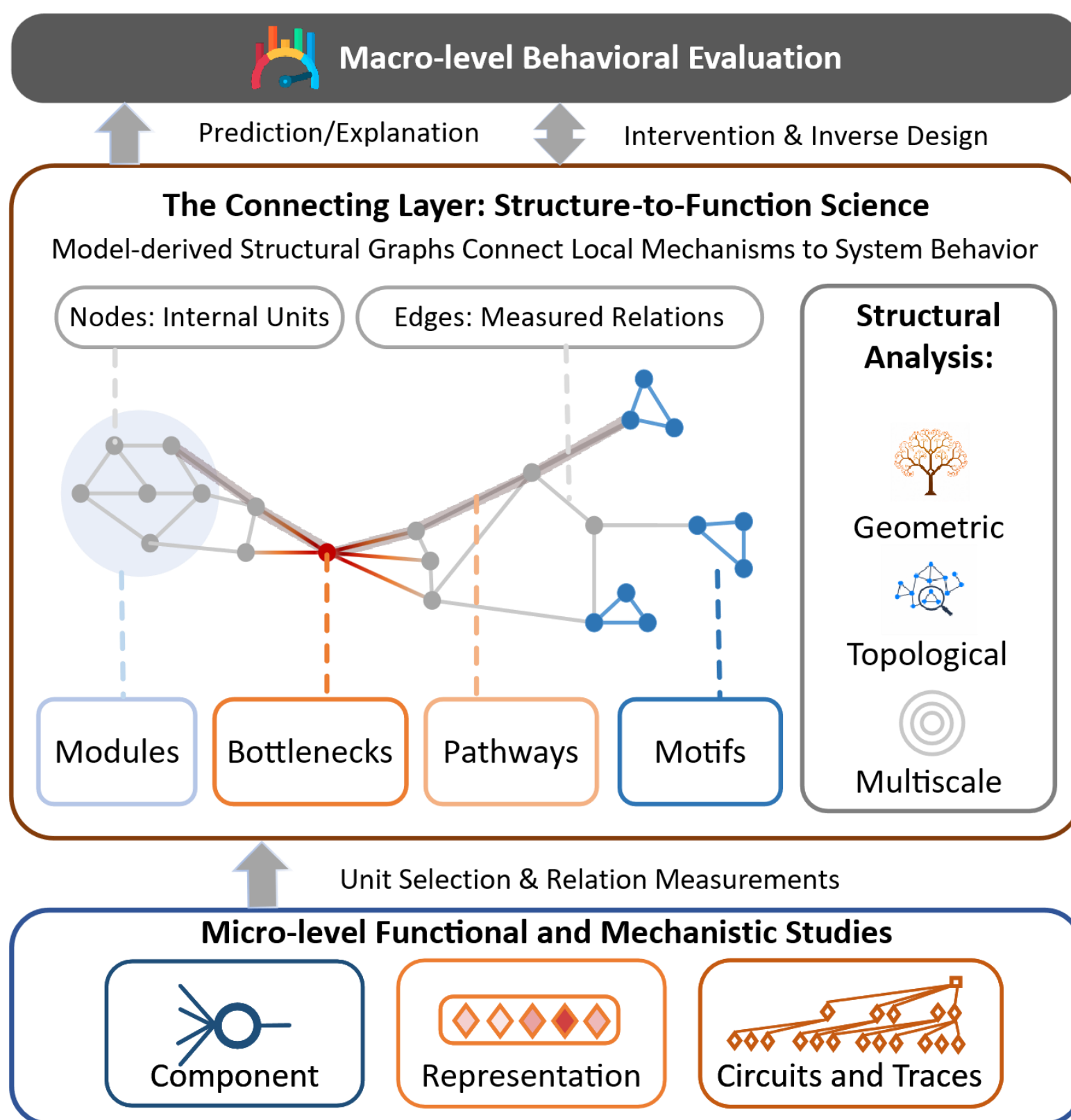
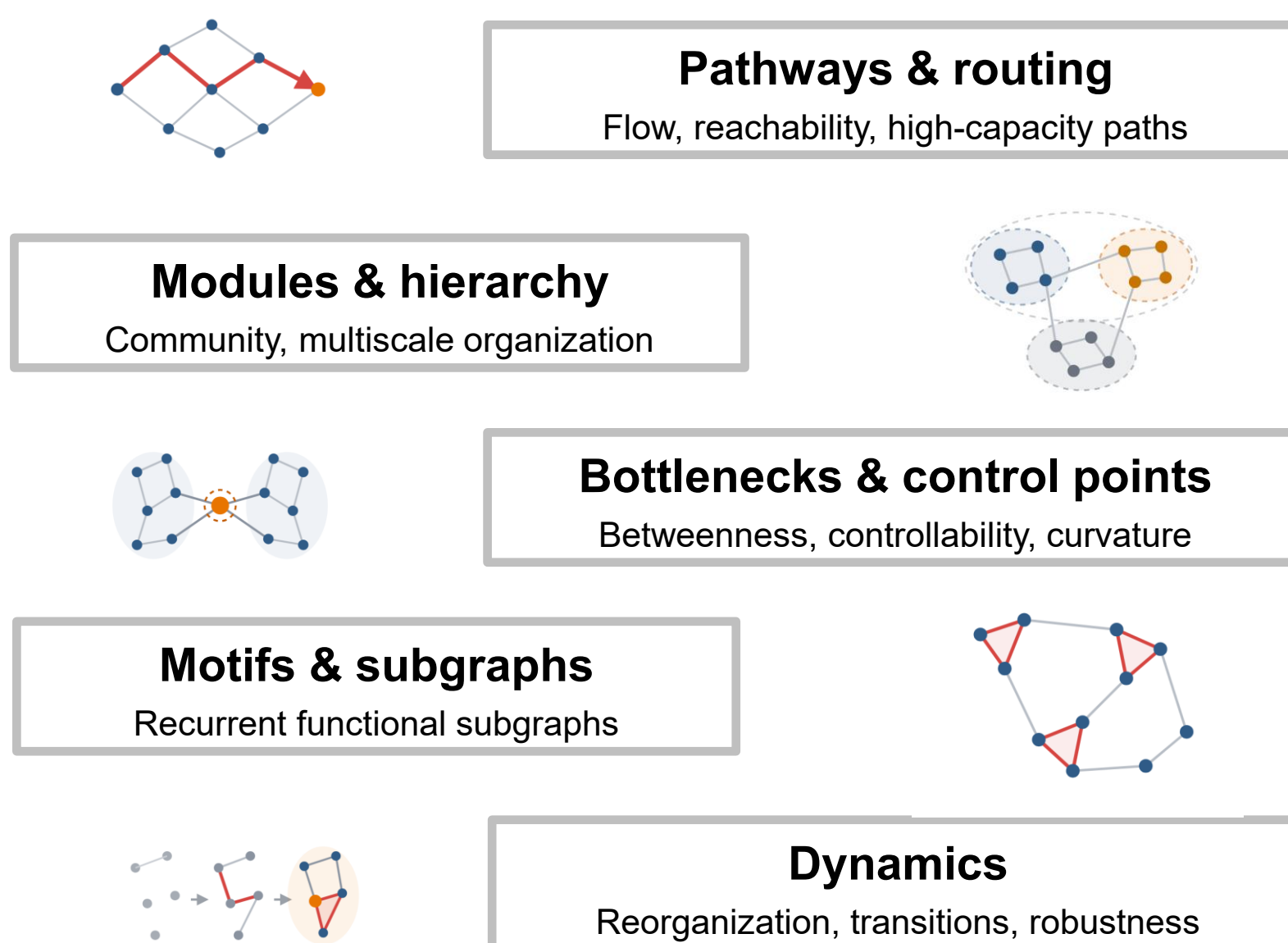


Figure 1: Structural organization links local mechanisms to system-level behavior.

Making Organization Measurable

Structural descriptors are **testable hypotheses** about internal organization.



When Is Structure Mechanistically Meaningful?

- **Recurrence → Prediction → Causal Intervention**
across contexts · Predicts held-out behavior · Produces specific effects vs. matched controls

$$\Delta B = B(M_i; do(S \leftarrow S'), c) - B(M_i; c) \quad (3)$$

Early Evidence and Hypothesis

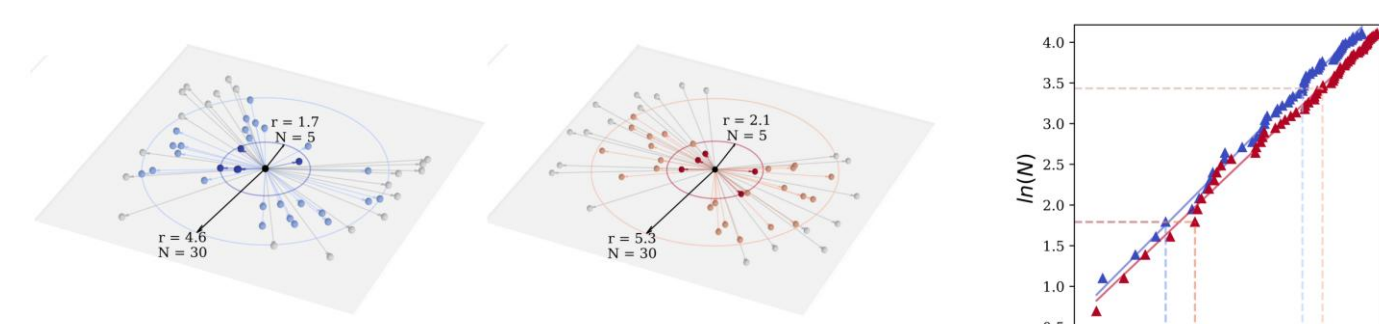


Figure 2: Fractality observed in LLM graphs.

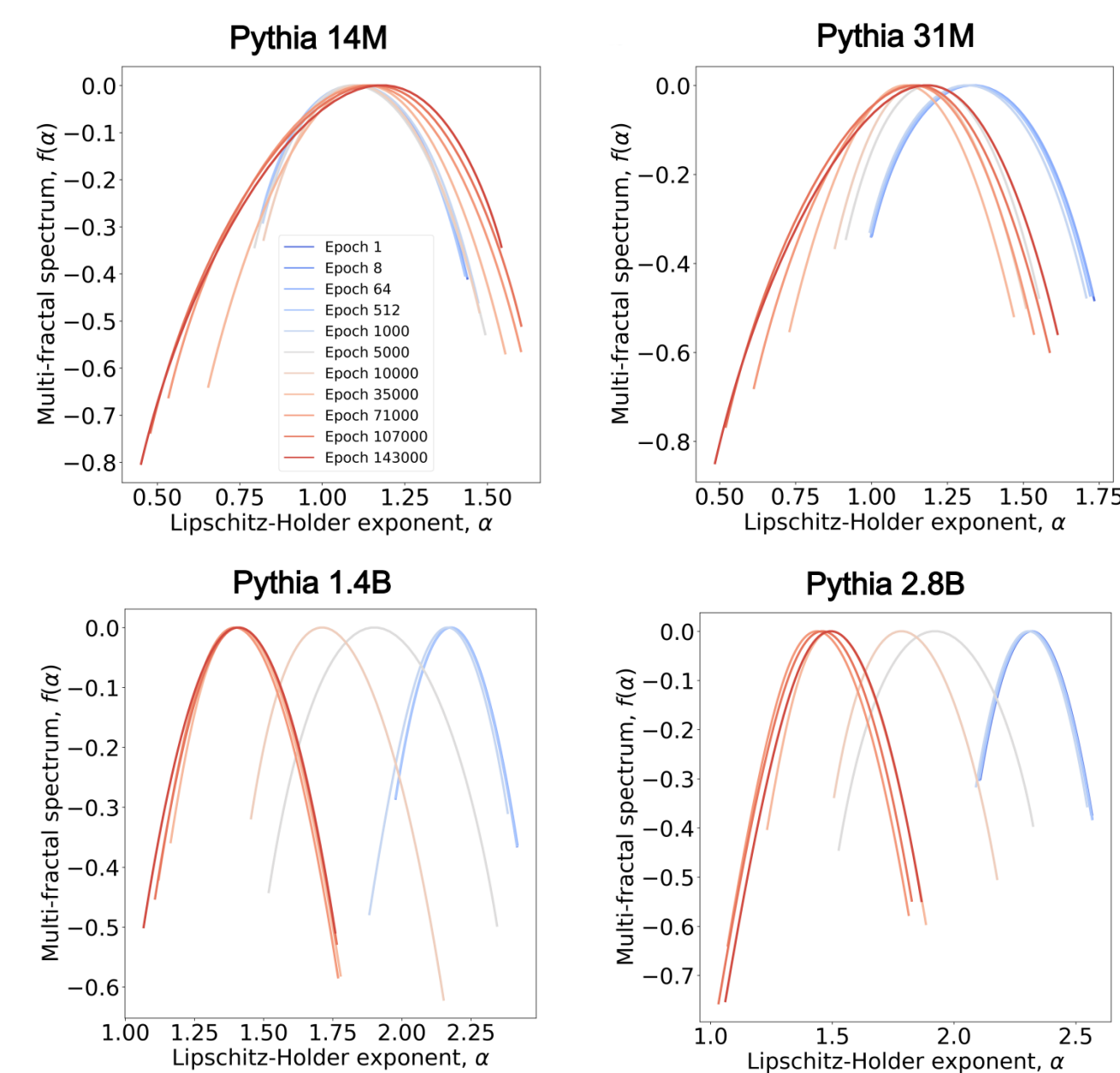


Figure 3: Internal multifractal structural signatures evolve systematically with model scale and training.

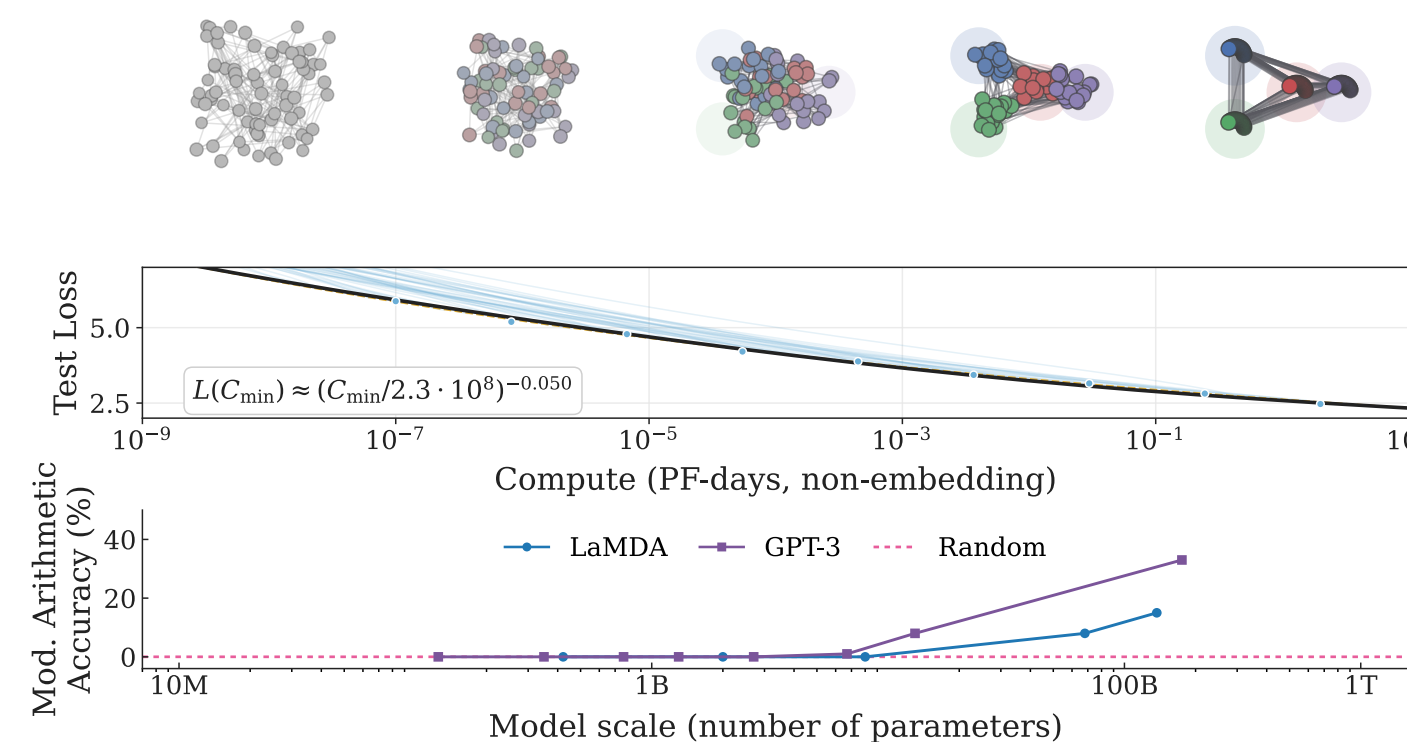


Figure 4: From external scale to internal organization

Key Scientific Questions

- Does a meaningful mesoscopic layer exist?
- How do structures form and reorganize?
- What scales internally when models scale externally?

SELECTED REFERENCES

[1] Bullmore & Sporns (2009). Complex brain networks: graph theoretical analysis of structural and functional systems. Nat. Rev. Neurosci. [2] Xiao et al. (2025). Neuron-based multifractal analysis of neuron interaction dynamics in large models. ICLR. [3] Conmy et al. (2023). Towards automated circuit discovery for mechanistic interpretability. NeurIPS. [4] Marks et al. (2025). Sparse feature circuits: discovering and editing interpretable causal graphs in language models. ICLR. [5] Lindsey et al. (2025). On the biology of a large language model. Transformer Circuits Thread.

Neuron-MFA Code:



Structure-to-Function Science: measure organization · predict behavior · intervene on structure